

Dataset 1: *Chlorella* light proteomics

Original Article: [Acclimation strategies of the green alga *Chlorella vulgaris* to different light regimes revealed by physiologic and comparative proteomic analysis](#)

Article Abstract

Acclimation to different light regimes is at the basis of survival for photosynthetic organisms, regardless of their evolutionary origin. Previous research efforts largely focused on acclimation events occurring at the level of the photosynthetic apparatus and often highlighted species-specific mechanisms. Here, we investigated the consequences of acclimation to different irradiances in *Chlorella vulgaris*, a green alga that is one of the most promising species for industrial application, focusing on both photosynthetic and mitochondrial activities. Moreover, proteomic analysis of cells acclimated to high light (HL) or low light (LL) allowed identification of the main targets of acclimation in terms of differentially expressed proteins. The results obtained demonstrate photosynthetic adaptation to HL versus LL that was only partially consistent with previous findings in *Chlamydomonas reinhardtii*, a model organism for green algae, but in many cases similar to vascular plant acclimation events. Increased mitochondrial respiration measured in HL-acclimated cells mainly relied on alternative oxidative pathways dissipating the excessive reducing power produced due to enhanced carbon flow. Finally, proteins involved in cell metabolism, intracellular transport, gene expression, and signaling—including a heliorhodopsin homolog—were identified as strongly differentially expressed in HL versus LL, suggesting their key roles in acclimation to different light regimes.

Sample Processing Protocol

Chlorella vulgaris cells were mechanically disrupted prior to a protein extraction step. Proteins were subsequently precipitated with acetone, resuspended, and sample protein amounts were measured utilizing BSA as standard. Proteins were then reduced, alkylated, digested with trypsin and purified. The resulting peptides were resuspended in LC-MS buffer prior to LC-MS analysis in directDIA mode.

Data Processing protocol

Raw files containing DIA data were downloaded from <https://www.ebi.ac.uk/pride/archive/projects/PXD037846> and processed with DIA-NN version 2.2.0 via command line, using a spectral library generated by DIA-NN in silico from the DIA data and the human FASTA genome. The shell command to execute the analysis is provided in the .txt file with the dataset. The dataset comprises 6 files : 3 files corresponding to cultivation of *Chlorella vulgaris* in high light conditions and 3 files corresponding to low light conditions. All outputs from DIA-NN are provided for exploration. The “Chlorella_light_proteomics_generated_speclib.parquet” file is the main DIA-NN output to be imported into QFeatures.

Dataset 2: Mitochondria after exercise proteomics

Original Article: [Sprint interval exercise disrupts mitochondrial ultrastructure driving a unique mitochondrial stress response and remodelling in men](#)

Article Abstract

Mitochondria play a central role in cellular metabolism and are crucial for maintaining cellular health. Exercise is a proven intervention to enhance mitochondrial function and prevent non-communicable diseases, yet the molecular mechanisms driving exercise-induced mitochondrial remodelling remain unclear. This study compares the effects of moderate-intensity continuous exercise (MICE) and sprint-interval exercise (SIE) on mitochondrial adaptations. We demonstrate that SIE induces mitochondrial stress and activates the mitochondrial unfolded protein response (UPR^{mt}), alongside disturbances in mitochondrial structure while MICE promotes mitochondrial adaptations characterized by increased mitochondrial content. Over an 8-week training period, moderate-intensity continuous training (MICT) and sprint-interval training (SIT) show divergent mitochondrial remodelling. Findings provide novel insights into the mechanisms by which different exercise prescriptions modulate mitochondrial signalling and remodelling in human skeletal muscle, offering a basis for personalized exercise strategies to optimize mitochondrial health.

Sample Processing Protocol

Skeletal muscle samples were solubilised with a probe sonicator with 30% amplitude in 5% SDS and 50 mM TEAB pH 8.5. Cycles of 10 seconds on and 10 seconds off were used for 1 minute. Protein concentration was assessed using the Pierce BCA Protein Assay Kit (Thermo Fisher Scientific) with 25 µg of each sample aliquoted in triplicates. 20 µg of whole cell pellets were prepared for mass spectrometry analysis with S-Trap mini spin columns (ProtiFi) as per manufacturer's instructions. Proteins were digested with trypsin (Thermo Fisher Scientific) at 1:10 trypsin to protein ratio. Further peptide purification was performed using SDB-RPS disc stage tips. Peptides were dried down using a CentriVap Benchtop Vacuum Concentrator (Labconco) and reconstituted in 45 µL 2% ACN, 0.1% trifluoroacetic acid (TFA) and 2 µL injected for liquid chromatography tandem mass spectrometry (LC-MS/MS). As per the described protocol above, a skeletal muscle library made up of 12 fractionated samples was generated for the search of data-independent acquisition (DIA) of the skeletal muscle samples.

Data Processing protocol

Raw files containing DIA data were downloaded from <https://www.ebi.ac.uk/pride/archive/projects/PXD059070> and processed with DIA-NN version 2.2.0 via command line, using a spectral library generated by DIA-NN in silico from the DIA data and the human FASTA genome. The shell command to execute the analysis is provided in the .txt file with the dataset. The dataset comprises 44 files from 24 paired samples from 12 volunteers before and after sprint interval training (SIT) and 20 paired samples from 10 volunteers before and after moderate intensity continuous training (MICT). All outputs from DIA-NN are provided for exploration. The "mitochondria_after_exercise_proteomics_speclib.parquet" files is the main DIA-NN output to be imported into QFeatures and "PRIDE_PXD059070_DIA_metadata.csv" is the metadata file to map sample names to conditions.